

B.E. Even Semester Midterm Examination, 2023-24(Even)

Subject: Environmental Science | **Code:** MC-HU 402

Q1. Goal and applications of environmental engineering.

Answer: Goal: Ensure societal development aligns with nature. Protect humans from adverse environmental effects and protect the environment from human/industrial damage. **Applications:** Water Supply/Treatment, Wastewater Management, Air Pollution Control, Solid Waste Management.

Q2. Components of ecosystem and role of EIA.

Answer: Components: Abiotic (non-living like soil, water) and Biotic (living like producers, consumers, decomposers). **EIA Role:** Evaluates environmental impacts before projects begin. Proposes preventive measures, corrective actions, and compensatory measures for sustainable development.

Q3. Explain nutrient cycle of an ecosystem.

Answer: Nutrient cycle (biogeochemical cycle) is the pathway where vital elements (Carbon, Nitrogen) move through biotic and abiotic components. Producers absorb nutrients -> Consumers eat plants -> Decomposers break down dead matter -> Nutrients return to the soil/air.

Q4. Define by-products. Sources, impact, control of soil pollution.

Answer: By-product: A secondary, incidental product derived from a manufacturing process (e.g., molasses). **Soil Pollution:** * Sources: Industrial waste, pesticides/fertilizers, solid waste. * Impacts: Reduces fertility, contaminates groundwater, biomagnification. * Control: Organic farming, effluent treatment, bioremediation.

Q5. Describe energy flow model with laws of thermodynamics.

Answer: Unidirectional flow governed by: 1. First Law (Conservation): Solar energy converts to chemical energy, transferred up the food chain. Total energy remains constant. 2. Second Law (Entropy): 10% Rule. Energy transfer is inefficient; ~90% lost as heat at each trophic level.